Request for Information (RFI)
National Oceanic and Atmospheric Administration (NOAA)
Office of Marine and Aviation Operations (OMAO)
Platform and Infrastructure Acquisition Division (PIAD)
Fisheries and Coastal Science Uncrewed Marine Systems

Disclaimer

This announcement constitutes a Request for Information (RFI) only for the purpose of determining market capability of sources and obtaining information. It does not constitute a Request for Proposals (RFP), a Request for Quotation (RFQ), or an indication that the Government will contract for any of the items and/or services discussed in this notice. Any formal solicitation that may subsequently be issued will be announced separately through the Government-wide Point of Entry (GPA) SAM.gov. Submission of a response to this RFI is entirely voluntary. The voluntary submission of a response to this RFI shall not obligate NOAA to pay or entitle the submitter of information to claim any direct or indirect costs or charges or any other remuneration whatsoever.

All information received will be safeguarded from unauthorized disclosure. Please do not submit any proprietary or classified information.

Purpose

NOAA is conducting market research to identify commercially available Uncrewed Marine System (UMS) solutions capable of supporting Fisheries and Coastal Science missions. The purpose of this RFI is to obtain information from industry regarding existing, commercially available, or near-commercial uncrewed marine surface vessel solutions capable of performing all or part of the requirements outlined below. NOAA hopes to use this information to accelerate the integration of emerging technologies and capabilities within the current observational at sea platforms.

Introduction

NOAA OMAO is one of the significant operational arms of NOAA, conducting vital mission work via marine and aircraft operations. Through its headquarters office in Silver Spring, MD, OMAO operates, maintains, and manages critical vessel acquisition programs to meet NOAA's prioritized National Marine Fisheries Service requirements.

Background

NOAA will be expanding its uncrewed systems capabilities within its current fleet to enhance its capabilities to perform its missions. NOAA is seeking state-of-the-art systems that are highly efficient to maximize its mission effectiveness.

NOAA has statutory responsibilities under the:

Magnuson–Stevens Fishery Conservation and Management Act
Endangered Species Act
Marine Mammal Protection Act

In support of these authorities, NOAA conducts fisheries surveys, habitat mapping, protected species assessments, and ecosystem-based research across U.S. Oceans and Great Lakes waters.

UMS Requirements and Characteristics

The Fisheries/Coastal Science UMS platform(s) have the potential to support living marine resource assessments, and environmental monitoring.

Respondents may address full-system solutions or modular approaches capable of integration.

1. **Mission Requirements:**

The following are notional requirements for Fisheries/Coastal Science UMS platforms, subject to change:

- **Assessment and Management of Living Marine Resources (LMR):** NOAA is responsible for the stewardship of the Nation's Ocean and Great Lakes resources and ecosystems. NOAA uses at-sea data to identify, characterize, monitor, and evaluate living marine resources and their ecosystems through physical, chemical, and biological observations.
- **Fisheries Science and Management:** Under the Magnuson Stevens Fisheries Conservation and Management Act, NOAA manages marine fisheries, including aquaculture, using the best available science resulting in sustainable fisheries harvest and production, rebuilding of fish stocks, conservation, and restoration of essential fish habitats and other support for fishing businesses and communities.
- **Protected Resources, Science and Management:** As required by the Endangered Species Act and the Marine Mammal Protection Act, NOAA assesses threatened and endangered marine and anadromous species and the ecosystems that sustain them to develop and implement best practices and conservation actions to reduce threats to protected species and their marine and coastal ecosystems.
- **Habitat Conservation and Restoration:** NOAA protects and restores habitat to sustain commercial and recreational fisheries, recover protected species, and maintain resilient marine ecosystems and coastal communities.
- **Ecosystem Studies:** NOAA assesses the status and trends of numerous ecosystems in United States waters, as well as the drivers of change, both within and outside marine protected areas and ranging in scale from specific locations (coral reefs or estuaries) to large marine ecosystems. Integrated ecosystem assessments are a science-based decision support process that provides the analytical framework to implement ecosystem-based approaches to manage our marine resources, including fisheries and protected species. These assessments synthesize data on physical, chemical, ecological, and human processes within ecosystems to provide sound interdisciplinary ecosystem-based science, which allow managers to evaluate trade-offs and ensure sustainable management of our Nation's resources.

The Fisheries/Coastal Science UMS platform(s) should demonstrate capability to perform or support:

- Ocean and Coastal biomass and bottom mapping using sonars to collect water column imagery and meet bottom mapping requirements.
- Water column profiling (conductivity, temperature, depth [CTD] Data Collection).
- Monitoring environmental parameters for ocean acidification and carbon cycle studies via underway pCO_2 measurements.
- Surface, midwater, and seafloor parameter sampling with scientific instruments.

Fisheries/Coastal Science (UMS) platforms should be able to provide Emergency Response capabilities to natural and man-made disasters, providing data collection for federal, state, and local agencies to minimize environmental and economic impacts from oil spills, chemical spills, vessel groundings, hazardous waste releases, and national security events.

2. Operating Profile and Performance Requirements

Respondents should demonstrate capabilities via remote or uncrewed systems to address system capabilities of the following desired operational profile. Platforms maybe able to be operated solely from a shore based communication and control system. NOAA welcomes all platforms that could meet some or all of the characteristics below and should note which can be accomplished.

- Days at Sea: 235 days per year
- Mission Duration: Up to 21 continuous days
- Sea State: Fully mission capable through Sea State 4
- Transit Speed: Minimum 10 knots
- Range: 5,500 nautical miles
- Endurance: 21 days continuous operation
- Operating Environment:
    - Minimum Air Temperature: -13°F
    - Minimum Seawater Temperature: 28°F
    - Maximum Air Temperature: 104°F (Dry Bulb), 90°F (Wet Bulb)
    - Maximum Seawater Temperature: 95°F

3. Mission and Scientific Capabilities

a. Acoustic Systems

The system may:

- Acquire high-resolution bathymetry
- Collect acoustic backscatter
- Collect water column imagery
- Support biomass assessments
- Enable high-precision acoustic navigation

Designs must minimize bubble sweepdown and other interference affecting sonar performance across the operating envelope.

b. Underwater Radiated Noise

Respondents may describe:

- Underwater radiated noise performance
- Applicable standards or radiated noise curves met
- Design features used to mitigate acoustic signature

c. Communications and Navigation

The system may include:

- Reliable, regulatory-compliant communications for unrestricted ocean service

- Integrated navigation systems
- Redundancy for critical systems
- Secure communications capability (if available)
- Ability to accommodate future technology insertion

d. Maintenance, Logistics, and Supportability

Respondents may describe:

- Organic vs. contractor logistics support concepts
- Maintenance intervals and requirements
- Shoreside infrastructure needs
- Availability of U.S.-based support
- Supply chain resilience

Information Sought

NOAA is seeking information on bubble sweepdown related to:

1. Company name, address, and point of contact
2. Description of existing commercial UMS platform(s)
3. Option for a Government Owned Contractor Operated (GOCO) arrangement
4. System maturity level (commercially available, in production, prototype, etc.)
5. Technical specifications (size, displacement, propulsion type, endurance, etc.)
6. Payload capacity and integration approach
7. Performance data relative to stated requirements
8. Underwater radiated noise data (if available)
9. Estimated cost range (Rough Order of Magnitude)
10. Production capacity and delivery timeline

Vendors may identify which requirements are fully met, partially met, or require modification.

Response Requirements

Responses shall clearly mark any proprietary information that is contained within them. Responses shall not contain classified information. RFI responses should be submitted electronically in a pdf format to the point of contact in this notice.

Please include the following with your response:

1. The RFI number stated herein.
2. Company name, division, mailing address, email address, and website (if applicable).
3. Point of contact, title, phone number, and email address.
4. Unique Entity Identifier (UEI) and confirmation of System for Award Management (SAM) registration.
5. Indicate what North American Industry Classification System (NAICS) codes your firm qualifies as a small business under.
6. State your firm's socio-economic status, if qualified as one or more of the following:
   ○ 8(a) firm (must be certified by the Small Business Administration [SBA])
   ○ Small Disadvantaged Business

○ Woman-Owned Small Business (must be certified by SBA)
○ HUBZone firm (must be certified by SBA)
○ Service-Disabled Veteran-Owned Small Business (must be certified by SBA)

7. Please provide any relevant GSA Schedule contracts, government-wide acquisition contracts (GWACs), Best in Class (BIC) contracts (e.g., 8(a) STARS II, Alliant 2, VETS, NITAAC), or other applicable IDIQs your firm holds across the Government, including the corresponding schedule or contract numbers.
8. Descriptive literature regarding capabilities and successful performance including any previous GOCO arrangements.
9. Be no more than 25 pages

NOTE: If necessary and required based on responses to this RFI, the Government reserves the right to schedule vendor meetings and have discussions between vendor personnel and NOAA representatives regarding the capabilities of the vendor supplies/services being suggested. All vendor meetings will be authorized, approved, and conducted by the Contracting Officer to ensure they do not create a potential exclusive situation that gives competitive advantage or otherwise disadvantages participants in any potential future competition.